

# Shutdown Unit Resets Unexpectedly

## Symptoms

- The SDU410 unit acknowledges alarms “on its own” with user interaction.

## How To Use This Tree

This symptom tree can be used to troubleshoot a malfunction. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

## Shoptalk

## Troubleshooting Summary

| STEPS   |                                                                                                | SPECIFICATIONS     |
|---------|------------------------------------------------------------------------------------------------|--------------------|
| STEP 1. | Check customer interface box                                                                   |                    |
|         | STEP 1A. Check the DCU410 unit or remote panel for flashing LED.                               | LED flashing?      |
|         | STEP 1B. Check the low speed oil pressure signal and return wires for an open.                 | Less than 10 ohms? |
|         | STEP 1B-1. Check the low speed oil pressure signal and return wires for a wire-to-wire short.  | Less than 10 ohms? |
|         | STEP 1B-2. Check the low speed oil pressure signal wire for a short to ground.                 | Less than 10 ohms? |
|         | STEP 1C. Check the high speed oil pressure signal and return wires for an open.                | Less than 10 ohms? |
|         | STEP 1C-1. Check the high speed oil pressure signal and return wires for a wire-to-wire short. | Less than 10 ohms? |

| STEPS |                                                                                                   | SPECIFICATIONS     |
|-------|---------------------------------------------------------------------------------------------------|--------------------|
|       | <b>STEP 1C-2.</b> Check the high speed oil pressure signal wire for a short to ground.            | Less than 10 ohms? |
|       | <b>STEP 1D.</b> Check the coolant pressure signal and return wires for an open.                   | Less than 10 ohms? |
|       | <b>STEP 1D-1.</b> Check the coolant pressure signal and return wires for a wire-to-wire short.    | Less than 10 ohms? |
|       | <b>STEP 1D-2.</b> Check the coolant pressure signal wire for a short to ground.                   | Less than 10 ohms? |
|       | <b>STEP 1E.</b> Check the coolant temperature signal and return wires for an open.                | Less than 10 ohms? |
|       | <b>STEP 1E-1.</b> Check the coolant temperature signal and return wires for a wire-to-wire short. | Less than 10 ohms? |
|       | <b>STEP 1E-2.</b> Check the coolant temperature signal wire for a short to ground.                | Less than 10 ohms? |
|       | <b>STEP 1F.</b> Check the engine speed 1 signal and return wires for an open.                     | Less than 10 ohms? |
|       | <b>STEP 1F-1.</b> Check the engine speed 1 signal and return wires for a wire-to-wire short.      | Less than 10 ohms? |
|       | <b>STEP 1F-2.</b> Check the engine speed 1 signal wire for a short to ground.                     | Less than 10 ohms? |
|       | <b>STEP 1G.</b> Check the engine speed 2 signal and return wires for an open.                     | Less than 10 ohms? |
|       | <b>STEP 1G-1.</b> Check the engine speed 2 signal and return wires for a wire-to-wire short.      | Less than 10 ohms? |
|       | <b>STEP 1G-2.</b> Check the engine speed 2 signal wire for a short to ground.                     | Less than 10 ohms? |

| STEPS          |                                                                                                                        | SPECIFICATIONS     |
|----------------|------------------------------------------------------------------------------------------------------------------------|--------------------|
|                | <b>STEP 1H.</b> Check the remote engine stop signal and return wires for an open.                                      | Less than 10 ohms? |
|                | <b>STEP 1H-1.</b> Check the remote engine stop signal and return wires for a wire-to-wire short.                       | Less than 10 ohms? |
|                | <b>STEP 1H-2.</b> Check the remote engine stop signal wire for a short to ground.                                      | Less than 10 ohms? |
|                | <b>STEP 1I.</b> Check the shutdown unit Modicon™ communication bus supply and return wires for an open.                | Less than 10 ohms? |
|                | <b>STEP 1I-1.</b> Check the shutdown unit Modicon™ communication bus supply and return wires for a wire-to-wire short. | Less than 10 ohms? |
|                | <b>STEP 1I-2.</b> Check the shutdown unit Modicon™ communication bus supply wire for a short to ground.                | Less than 10 ohms? |
| <b>STEP 2.</b> | Check the OEM wiring harness.                                                                                          |                    |
|                | <b>STEP 2A.</b> Check the low speed oil pressure signal and return wires for an open.                                  | Less than 10 ohms? |
|                | <b>STEP 2A-1.</b> Check the low speed oil pressure signal and return wires for a wire-to-wire short.                   | Less than 10 ohms? |
|                | <b>STEP 2B.</b> Check the high speed oil pressure signal and return wires for an open.                                 | Less than 10 ohms? |
|                | <b>STEP 2B-1.</b> Check the high speed oil pressure signal and return wires for a wire-to-wire short.                  | Less than 10 ohms? |
|                | <b>STEP 2C.</b> Check the coolant pressure signal and return wires for an open.                                        | Less than 10 ohms? |

| STEPS |                                                                                                   | SPECIFICATIONS     |
|-------|---------------------------------------------------------------------------------------------------|--------------------|
|       | <b>STEP 2C-1.</b> Check the coolant pressure signal and return wires for a wire-to-wire short.    | Less than 10 ohms? |
|       | <b>STEP 2D.</b> Check the coolant temperature signal and return wires for an open.                | Less than 10 ohms? |
|       | <b>STEP 2D-1.</b> Check the coolant temperature signal and return wires for a wire-to-wire short. | Less than 10 ohms? |
|       | <b>STEP 2E.</b> Check the engine speed 1 signal and return wires for an open.                     | Less than 10 ohms? |
|       | <b>STEP 2E-1.</b> Check the engine speed 1 signal and return wires for a wire-to-wire short.      | Less than 10 ohms? |
|       | <b>STEP 2F.</b> Check the engine speed 2 signal and return wires for an open.                     | Less than 10 ohms? |
|       | <b>STEP 2F-1.</b> Check the engine speed 2 signal and return wires for a wire-to-wire short.      | Less than 10 ohms? |

## STEP 1. Check the customer interface box.

### STEP 1A. Check the DCU410 unit for alarms and LED illumination.

#### Conditions :

- Check for alarm and LED illumination

| Action | Specification/Repair | Next Step |
|--------|----------------------|-----------|
|        |                      |           |

|                                                                                                                                                                                                                    |                                                                                                                                                              |                        |
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| <p>Check the SDU410 unit Acknowledge button.</p> <p>NOTE: Be sure the Acknowledge button is <b>not</b> stuck in the activation mode. Press the Acknowledge button to see if it changes to steady illumination.</p> | <p>Acknowledge button stuck?</p> <p><b>YES</b></p> <p><b>Repair :</b></p> <p>Replace the SDU410 unit.<br/>Contact a Cummins® Authorized Repair Location.</p> | <b>Repair complete</b> |
|                                                                                                                                                                                                                    | <p>Acknowledge button stuck?</p> <p><b>NO</b></p>                                                                                                            | <b>1B</b>              |

### STEP 1B. Check the low speed oil pressure signal and return wires for an open.

#### Conditions :

- Open the customer interface box.
- Disconnect the low speed oil pressure signal and return wires at the SDU410 unit and C4 connector.

| Action                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Specification/Repair                                                                                                                                                                                                    | Next Step              |
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| <p>Check the signal and return wires for an open.</p> <ul style="list-style-type: none"> <li>• Place one test lead on the low speed oil pressure signal wire at the SDU410 unit.</li> <li>• Place the other test lead on the low speed oil pressure signal pin at the C4 connector.</li> <li>• Place one test lead on the low speed oil pressure return wire at the SDU410 unit.</li> <li>• Place the other test lead on the low speed oil pressure return pin at the C4 connector.</li> </ul> <p>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</p> | <p>Less than 10 ohms?</p> <p><b>YES</b></p>                                                                                                                                                                             | <b>1B-1</b>            |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <p>Less than 10 ohms?</p> <p><b>NO</b></p> <p><b>Repair :</b></p> <p>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15.<br/>(/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</p> | <b>Repair complete</b> |

**STEP 1B-1. Check the low speed oil pressure signal and return wires for a wire-to-wire short.**

**Conditions :**

- Open the customer interface box.
- Disconnect the low speed oil pressure signal and return wires at the SDU410 unit and C4 connector.

| Action                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Specification/Repair                                                                                                                                                                                         | Next Step              |
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| Check the low speed oil pressure signal and return wires for a wire-to-wire short. <ul style="list-style-type: none"><li>• Place one test lead on the low speed oil pressure signal wire at the SDU410 unit.</li><li>• Place the other test lead on all other pins at the C4 connector.</li><li>• Place one test lead on the low speed oil pressure return wire at the SDU410 unit.</li><li>• Place the other test lead on all other pins at the C4 connector.</li></ul> Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification. | Less than 10 ohms?<br><b>YES</b><br><br><b>Repair :</b><br><br>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15.<br>(/qs3/pubsys2/xml/en/procedures/116/116-015-023.html) | <b>Repair complete</b> |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Less than 10 ohms?<br><b>NO</b>                                                                                                                                                                              | <b>1B-2</b>            |

**STEP 1B-2. Check the low speed oil pressure signal wire for a short to ground.**

**Conditions :**

- Open the customer interface box.
- Disconnect the low speed oil pressure signal wire at the SDU410 unit and C4 connector.

| Action | Specification/Repair | Next Step |
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| <p>Check the low speed oil pressure signal wire for a short to ground.</p> <ul style="list-style-type: none"> <li>Place one test lead on the low speed oil pressure signal wire at the SDU410 unit.</li> <li>Place the other test lead on panel ground.</li> </ul> <p>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</p> | <p>Less than 10 ohms?</p> <p><b>YES</b></p> <p><b>Repair :</b></p> <p>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15.<br/>(/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</p> | <p><b>Repair complete</b></p> |
|                                                                                                                                                                                                                                                                                                                                                                       | <p>Less than 10 ohms?</p> <p><b>NO</b></p>                                                                                                                                                                               | <p><b>1C</b></p>              |

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| <p><b>STEP 1C. Check the high speed oil pressure signal and return wires for an open.</b></p>                                                                                                                              |                             |                  |
| <p><b>Conditions :</b></p> <ul style="list-style-type: none"> <li>Open the customer interface box.</li> <li>Disconnect the high speed oil pressure signal and return wires at the SDU410 unit and connector C4.</li> </ul> |                             |                  |
| <b>Action</b>                                                                                                                                                                                                              | <b>Specification/Repair</b> | <b>Next Step</b> |

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| <p>Check the high speed oil pressure signal and return wires for an open.</p> <ul style="list-style-type: none"> <li>Place one test lead on the high speed oil pressure signal wire at the SDU410 unit.</li> <li>Place the other test lead on the high speed oil pressure signal pin at the C4 connector.</li> <li>Place one test lead on the high speed oil pressure return wire at the SDU410 unit.</li> <li>Place the other test lead on the high speed oil pressure return pin at the C4 connector.</li> </ul> <p>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</p> | <p>Less than 10 ohms?</p> <p><b>YES</b></p>                                                                                                                                                                             | <p><b>1C-1</b></p>            |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <p>Less than 10 ohms?</p> <p><b>NO</b></p> <p><b>Repair :</b></p> <p>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15.<br/>(/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</p> | <p><b>Repair complete</b></p> |

**STEP 1C-1. Check the high speed oil pressure signal and return wires for a wire-to-wire short.**

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| <p><b>Conditions :</b></p> <ul style="list-style-type: none"> <li>Open the customer interface box.</li> <li>Disconnect the high speed oil pressure signal and return wires at the SDU410 unit and C4 connector.</li> </ul> |                             |                  |
| <b>Action</b>                                                                                                                                                                                                              | <b>Specification/Repair</b> | <b>Next Step</b> |



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| <p>Check the high speed oil pressure signal and return wires for a wire-to-wire short.</p> <ul style="list-style-type: none"> <li>Place one test lead on the high speed oil pressure signal wire at the SDU410 unit.</li> <li>Place the other test lead on all other pins at the C4 connector.</li> <li>Place one test lead on the high speed oil pressure return wire at the SDU410 unit.</li> <li>Place the other test lead on all other pins at the C4 connector.</li> </ul> <p>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</p> | <p>Less than 10 ohms?</p> <p><b>YES</b></p> <p><b>Repair :</b></p> <p>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15.<br/>(/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</p> | <p><b>Repair complete</b></p> |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <p>Less than 10 ohms?</p> <p><b>NO</b></p>                                                                                                                                                                               | <p><b>1C-2</b></p>            |

## STEP 1C-2. Check the low speed oil pressure signal wire for a short to ground.

| <p><b>Conditions :</b></p> <ul style="list-style-type: none"> <li>Open the customer interface box.</li> <li>Disconnect the low speed oil pressure signal wire at the SDU410 unit and C4 connector.</li> </ul> |                      |           |
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| Action                                                                                                                                                                                                        | Specification/Repair | Next Step |

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| <p>Check the high speed oil pressure signal wire for a short to ground.</p> <ul style="list-style-type: none"> <li>Place one test lead on the low speed oil pressure signal wire at the SDU410 unit.</li> <li>Place the other test lead on panel ground.</li> </ul> <p>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</p> | <p>Less than 10 ohms?</p> <p><b>YES</b></p> <p><b>Repair :</b></p> <p>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15.<br/>(/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</p> | <p><b>Repair complete</b></p> |
|                                                                                                                                                                                                                                                                                                                                                                        | <p>Less than 10 ohms?</p> <p><b>NO</b></p>                                                                                                                                                                               | <p><b>1D</b></p>              |

### STEP 1D. Check the coolant pressure signal and return wires for an open.

| <p><b>Conditions :</b></p> <ul style="list-style-type: none"> <li>Open the customer interface box.</li> <li>Disconnect the coolant pressure signal and return wires at the SDU410 unit and connector C4.</li> </ul> |                      |           |
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| Action                                                                                                                                                                                                              | Specification/Repair | Next Step |

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| <p>Check the coolant pressure signal and return wires for an open.</p> <ul style="list-style-type: none"> <li>Place one test lead on the coolant pressure signal wire at the SDU410 unit.</li> <li>Place the other test lead on the coolant pressure signal pin at the C4 connector.</li> <li>Place one test lead on the coolant pressure return wire at the SDU410 unit.</li> <li>Place the other test lead on the coolant pressure return pin at the C4 connector.</li> </ul> <p>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</p> | <p>Less than 10 ohms?</p> <p><b>YES</b></p>                                                                                                                                                                             | <p><b>1D-1</b></p>            |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <p>Less than 10 ohms?</p> <p><b>NO</b></p> <p><b>Repair :</b></p> <p>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15.<br/>(/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</p> | <p><b>Repair complete</b></p> |

**STEP 1D-1. Check the coolant pressure signal and return wires for a wire-to-wire short.**

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| <p><b>Conditions :</b></p> <ul style="list-style-type: none"> <li>Open the customer interface box.</li> <li>Disconnect the coolant pressure signal and return wires at the SDU410 unit and C4 connector.</li> </ul> |                             |                  |
| <b>Action</b>                                                                                                                                                                                                       | <b>Specification/Repair</b> | <b>Next Step</b> |

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| <p>Check the coolant pressure signal and return wires for a wire-to-wire short.</p> <ul style="list-style-type: none"> <li>• Place one test lead on the coolant pressure signal wire at the SDU410 unit.</li> <li>• Place the other test lead on all other pins at the C4 connector.</li> <li>• Place one test lead on the coolant pressure return wire at the SDU410 unit.</li> <li>• Place the other test lead on all other pins at the C4 connector.</li> </ul> <p>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</p> | <p>Less than 10 ohms?</p> <p><b>YES</b></p> <p><b>Repair :</b></p> <p>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15.<br/>(/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</p> | <p><b>Repair complete</b></p> |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <p>Less than 10 ohms?</p> <p><b>NO</b></p>                                                                                                                                                                               | <p><b>1D-2</b></p>            |

## STEP 1D-2. Check the coolant pressure signal wire for a short to ground.

### Conditions :

- Open the customer interface box.
- Disconnect the coolant pressure signal wire at the SDU410 unit and C4 connector.

| Action | Specification/Repair | Next Step |
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| <p>Check the coolant pressure signal wire for a short to ground.</p> <ul style="list-style-type: none"> <li>Place one test lead on the coolant pressure signal wire at the SDU410 unit.</li> <li>Place the other test lead on panel ground.</li> </ul> <p>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</p> | <p>Less than 10 ohms?</p> <p><b>YES</b></p> <p><b>Repair :</b></p> <p>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15.<br/>(/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</p> | <p><b>Repair complete</b></p> |
|                                                                                                                                                                                                                                                                                                                                                           | <p>Less than 10 ohms?</p> <p><b>NO</b></p>                                                                                                                                                                               | <p><b>1E</b></p>              |

### STEP 1E. Check the coolant temperature signal and return wires for an open.

| <p><b>Conditions :</b></p> <ul style="list-style-type: none"> <li>Open the customer interface box.</li> <li>Disconnect the coolant temperature signal and return wires at the SDU410 unit and connector C4.</li> </ul> |                      |           |
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| Action                                                                                                                                                                                                                 | Specification/Repair | Next Step |

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| <p>Check the coolant temperature signal and return wires for an open.</p> <ul style="list-style-type: none"> <li>Place one test lead on the coolant temperature signal wire at the SDU410 unit.</li> <li>Place the other test lead on the coolant temperature signal pin at the C4 connector.</li> <li>Place one test lead on the coolant temperature return wire at the SDU410 unit.</li> <li>Place the other test lead on the coolant temperature return pin at the C4 connector.</li> </ul> <p>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</p> | <p>Less than 10 ohms?</p> <p><b>YES</b></p>                                                                                                                                                                             | <p><b>1E-1</b></p>            |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <p>Less than 10 ohms?</p> <p><b>NO</b></p> <p><b>Repair :</b></p> <p>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15.<br/>(/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</p> | <p><b>Repair complete</b></p> |

**STEP 1E-1. Check the coolant temperature signal and return wires for a wire-to-wire short.**

**Conditions :**

- Open the customer interface box.
- Disconnect the coolant temperature signal and return wires at the SDU410 unit and C4 connector.

| Action | Specification/Repair | Next Step |
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| <p>Check the coolant temperature signal and return wires for a wire-to-wire short.</p> <ul style="list-style-type: none"> <li>Place one test lead on the coolant temperature signal wire at the SDU410 unit.</li> <li>Place the other test lead on all other pins at the C4 connector.</li> <li>Place one test lead on the coolant temperature return wire at the SDU410 unit.</li> <li>Place the other test lead on all other pins at the C4 connector.</li> </ul> <p>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</p> | <p>Less than 10 ohms?</p> <p><b>YES</b></p> <p><b>Repair :</b></p> <p>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15.<br/>(/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</p> | <p><b>Repair complete</b></p> |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <p>Less than 10 ohms?</p> <p><b>NO</b></p>                                                                                                                                                                               | <p><b>1E-2</b></p>            |

## STEP 1E-2. Check the coolant temperature signal wire for a short to ground.

### Conditions :

- Open the customer interface box.
- Disconnect the coolant temperature signal wire at the SDU410 unit and C4 connector.

| Action | Specification/Repair | Next Step |
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| <p>Check the coolant temperature signal wire for a short to ground.</p> <ul style="list-style-type: none"> <li>Place one test lead on the coolant temperature signal wire at the SDU410 unit.</li> <li>Place the other test lead on panel ground.</li> </ul> <p>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</p> | <p>Less than 10 ohms?</p> <p><b>YES</b></p> <p><b>Repair :</b></p> <p>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15.<br/>(/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</p> | <p><b>Repair complete</b></p> |
|                                                                                                                                                                                                                                                                                                                                                                 | <p>Less than 10 ohms?</p> <p><b>NO</b></p>                                                                                                                                                                               | <p><b>1F</b></p>              |

### STEP 1F. Check the engine speed 1 signal and return wires for an open.

| <p><b>Conditions :</b></p> <ul style="list-style-type: none"> <li>Open the customer interface box.</li> <li>Disconnect the engine speed 1 signal and return wires at the SDU410 unit and connector C4.</li> </ul> |                      |           |
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| Action                                                                                                                                                                                                            | Specification/Repair | Next Step |



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| <p>Check the engine speed 1 signal and return wires for an open.</p> <ul style="list-style-type: none"> <li>Place one test lead on the engine speed 1 signal wire at the SDU410 unit.</li> <li>Place the other test lead on the engine speed 1 signal pin at the C4 connector.</li> <li>Place one test lead on the engine speed 1 return wire at the SDU410 unit.</li> <li>Place the other test lead on the engine speed 1 return pin at the C4 connector.</li> </ul> <p>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</p> | <p>Less than 10 ohms?</p> <p><b>YES</b></p>                                                                                                                                                                             | <p><b>1F-1</b></p>            |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <p>Less than 10 ohms?</p> <p><b>NO</b></p> <p><b>Repair :</b></p> <p>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15.<br/>(/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</p> | <p><b>Repair complete</b></p> |

### STEP 1F-1. Check the engine speed 1 signal and return wires for a wire-to-wire short.

#### Conditions :

- Open the customer interface box.
- Disconnect the engine speed 1 signal and return wires at the SDU410 unit and C4 connector.

| Action | Specification/Repair | Next Step |
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                          |                               |
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| <p>Check the engine speed 1 signal and return wires for a wire-to-wire short.</p> <ul style="list-style-type: none"> <li>Place one test lead on the engine speed 1 signal wire at the SDU410 unit.</li> <li>Place the other test lead on all other pins at the C4 connector.</li> <li>Place one test lead on the engine speed 1 return wire at the SDU410 unit.</li> <li>Place the other test lead on all other pins at the C4 connector.</li> </ul> <p>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</p> | <p>Less than 10 ohms?</p> <p><b>YES</b></p> <p><b>Repair :</b></p> <p>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15.<br/>(/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</p> | <p><b>Repair complete</b></p> |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <p>Less than 10 ohms?</p> <p><b>NO</b></p>                                                                                                                                                                               | <p><b>1F-2</b></p>            |

## STEP 1F-2. Check the engine speed 1 signal wire for a short to ground.

### Conditions :

- Open the customer interface box.
- Disconnect the engine speed 1 signal wire at the SDU410 unit and C4 connector.

| Action | Specification/Repair | Next Step |
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| <p>Check the engine speed 1 signal wire for a short to ground.</p> <ul style="list-style-type: none"> <li>Place one test lead on the engine speed 1 signal wire at the SDU410 unit.</li> <li>Place the other test lead on panel ground.</li> </ul> <p>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</p> | <p>Less than 10 ohms?</p> <p><b>YES</b></p> <p><b>Repair :</b></p> <p>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15.<br/>(/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</p> | <p><b>Repair complete</b></p> |
|                                                                                                                                                                                                                                                                                                                                                       | <p>Less than 10 ohms?</p> <p><b>NO</b></p>                                                                                                                                                                               | <p><b>1G</b></p>              |

### STEP 1G. Check the engine speed 2 signal and return wires for an open.

| <p><b>Conditions :</b></p> <ul style="list-style-type: none"> <li>Open the customer interface box.</li> <li>Disconnect the engine speed 2 signal and return wires at the SDU410 unit and connector C4.</li> </ul> |                      |           |
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| Action                                                                                                                                                                                                            | Specification/Repair | Next Step |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                         |                               |
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| <p>Check the engine speed 2 signal and return wires for an open.</p> <ul style="list-style-type: none"> <li>Place one test lead on the engine speed 2 signal wire at the SDU410 unit.</li> <li>Place the other test lead on the engine speed 2 signal pin at the C4 connector.</li> <li>Place one test lead on the engine speed 2 return wire at the SDU410 unit.</li> <li>Place the other test lead on the engine speed 2 return pin at the C4 connector.</li> </ul> <p>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</p> | <p>Less than 10 ohms?</p> <p><b>YES</b></p>                                                                                                                                                                             | <p><b>1G-1</b></p>            |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <p>Less than 10 ohms?</p> <p><b>NO</b></p> <p><b>Repair :</b></p> <p>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15.<br/>(/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</p> | <p><b>Repair complete</b></p> |

### STEP 1G-1. Check the engine speed 2 signal and return wires for a wire-to-wire short.

#### Conditions :

- Open the customer interface box.
- Disconnect the engine speed 2 signal and return wires at the SDU410 unit and C4 connector.

| Action | Specification/Repair | Next Step |
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| <p>Check the engine speed 2 signal and return wires for a wire-to-wire short.</p> <ul style="list-style-type: none"> <li>Place one test lead on the engine speed 2 signal wire at the SDU410 unit.</li> <li>Place the other test lead on all other pins at the C4 connector.</li> <li>Place one test lead on the engine speed 2 return wire at the SDU410 unit.</li> <li>Place the other test lead on all other pins at the C4 connector.</li> </ul> <p>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</p> | <p>Less than 10 ohms?</p> <p><b>YES</b></p> <p><b>Repair :</b></p> <p>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15.<br/>(/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</p> | <p><b>Repair complete</b></p> |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <p>Less than 10 ohms?</p> <p><b>NO</b></p>                                                                                                                                                                               | <p><b>1G-2</b></p>            |

## STEP 1G-2. Check the engine speed 2 signal wire for a short to ground.

### Conditions :

- Open the customer interface box.
- Disconnect the engine speed 2 signal wire at the SDU410 unit and C4 connector.

| Action | Specification/Repair | Next Step |
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| <p>Check the engine speed 2 signal wire for a short to ground.</p> <ul style="list-style-type: none"> <li>Place one test lead on the engine speed 2 signal wire at the SDU410 unit.</li> <li>Place the other test lead on panel ground.</li> </ul> <p>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</p> | <p>Less than 10 ohms?</p> <p><b>YES</b></p> <p><b>Repair :</b></p> <p>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15.<br/>(/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</p> | <p><b>Repair complete</b></p> |
|                                                                                                                                                                                                                                                                                                                                                       | <p>Less than 10 ohms?</p> <p><b>NO</b></p>                                                                                                                                                                               | <p><b>1H</b></p>              |

### STEP 1H. Check the remote engine stop signal and return wires for an open.

| <p><b>Conditions :</b></p> <ul style="list-style-type: none"> <li>Open the customer interface box.</li> <li>Disconnect the remote engine stop signal and return wires at the SDU410 unit and X4 connector.</li> </ul> |                      |           |
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| Action                                                                                                                                                                                                                | Specification/Repair | Next Step |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                         |                               |
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| <p>Check the remote engine stop signal and return wires for an open.</p> <ul style="list-style-type: none"> <li>Place one test lead on the remote engine stop signal wire at the SDU410 unit.</li> <li>Place the other test lead on the remote engine stop signal pin at the X4 connector.</li> <li>Place one test lead on the remote engine stop return wire at the SDU410 unit.</li> <li>Place the other test lead on the remote engine stop return pin at the X4 connector.</li> </ul> <p>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</p> | <p>Less than 10 ohms?</p> <p><b>YES</b></p>                                                                                                                                                                             | <p><b>1H-1</b></p>            |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <p>Less than 10 ohms?</p> <p><b>NO</b></p> <p><b>Repair :</b></p> <p>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15.<br/>(/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</p> | <p><b>Repair complete</b></p> |

**STEP 1H-1. Check the remote engine stop signal and return wires for a wire-to-wire short.**

| <p><b>Conditions :</b></p> <ul style="list-style-type: none"> <li>Open the customer interface box.</li> <li>Disconnect the remote engine stop signal and return wires at the SDU410 unit and X4 connector.</li> </ul> |                      |           |
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| Action                                                                                                                                                                                                                | Specification/Repair | Next Step |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                          |                               |
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| <p>Check the remote engine stop signal and return wires for a wire-to-wire short.</p> <ul style="list-style-type: none"> <li>Place one test lead on the remote engine stop signal wire at the SDU410 unit.</li> <li>Place the other test lead on all other pins at the X4 connector.</li> <li>Place one test lead on the remote engine stop return wire at the SDU410 unit.</li> <li>Place the other test lead on all other pins at the X4 connector.</li> </ul> <p>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</p> | <p>Less than 10 ohms?</p> <p><b>YES</b></p> <p><b>Repair :</b></p> <p>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15.<br/>(/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</p> | <p><b>Repair complete</b></p> |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <p>Less than 10 ohms?</p> <p><b>NO</b></p>                                                                                                                                                                               | <p><b>1H-2</b></p>            |

## STEP 1H-2. Check the remote engine stop signal wire for a short to ground.

### Conditions :

- Open the customer interface box.
- Disconnect the remote engine stop signal wire at the SDU410 unit and X4 connector.

| Action | Specification/Repair | Next Step |
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|                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                          |                               |
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| <p>Check the remote engine stop signal wire for a short to ground.</p> <ul style="list-style-type: none"> <li>Place one test lead on the remote engine stop signal wire at the SDU410 unit.</li> <li>Place the other test lead on panel ground.</li> </ul> <p>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</p> | <p>Less than 10 ohms?</p> <p><b>YES</b></p> <p><b>Repair :</b></p> <p>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15.<br/>(/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</p> | <p><b>Repair complete</b></p> |
|                                                                                                                                                                                                                                                                                                                                                               | <p>Less than 10 ohms?</p> <p><b>NO</b></p>                                                                                                                                                                               | <p><b>11</b></p>              |

**STEP 11. Check the shutdown unit Modicon™ communication bus supply and return wires for an open.**

**Conditions :**

- Open the customer interface box.
- Disconnect the shutdown unit Modicon™ communication bus supply and return wires at the SDU410 unit and DCU410 unit.

| Action | Specification/Repair | Next Step |
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| <p>Check the shutdown unit Modicon™ communication bus supply and return wires for an open.</p> <ul style="list-style-type: none"> <li>Place one test lead on the shutdown unit Modicon™ communication bus supply wire at the SDU410 unit.</li> <li>Place the other test lead on the shutdown unit Modicon™ communication bus supply wire at the DCU410 unit.</li> <li>Place one test lead on the shutdown unit Modicon™ communication bus return wire at the SDU410 unit.</li> <li>Place the other test lead on the shutdown unit Modicon™ communication bus return wire at the DCU410 unit.</li> </ul> <p>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</p> | <p>Less than 10 ohms?</p> <p><b>YES</b></p>                                                                                                                                                                             | <p><b>11-1</b></p>            |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <p>Less than 10 ohms?</p> <p><b>NO</b></p> <p><b>Repair :</b></p> <p>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15.<br/>(/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</p> | <p><b>Repair complete</b></p> |

**STEP 11-1. Check the shutdown unit Modicon™ communication bus supply and return wires for a wire-to-wire short.**

**Conditions :**

- Open the customer interface box.
- Disconnect the shutdown unit Modicon™ communication bus supply and return wires at the SDU410 unit and DCU410 unit.

| Action | Specification/Repair | Next Step |
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| <p>Check the shutdown unit Modicon™ communication bus supply and return wires for a wire-to-wire short.</p> <ul style="list-style-type: none"> <li>• Place one test lead on the shutdown unit Modicon™ communication bus supply wire at the SDU410 unit.</li> <li>• Place the other test lead on all other wires at the DCU410 unit.</li> <li>• Place one test lead on the shutdown unit Modicon™ communication bus return wire at the SDU410 unit.</li> <li>• Place the other test lead on all other wires at the DCU410 unit.</li> </ul> | <p>Less than 10 ohms?</p> <p><b>YES</b></p> <p><b>Repair :</b></p> <p>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15.<br/>(/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</p> | <p><b>Repair complete</b></p> |
| <p>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                         | <p>Less than 10 ohms?</p> <p><b>NO</b></p>                                                                                                                                                                               | <p><b>11-2</b></p>            |

**STEP 11-2. Check the shutdown unit Modicon™ communication bus supply wire for a short to ground.**

**Conditions :**

- Open the customer interface box.
- Disconnect the shutdown unit Modicon™ communication bus supply wire at the SDU410 unit and DCU410 unit.

| Action | Specification/Repair | Next Step |
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| <p>Check the shutdown unit Modicon™ communication bus supply wire for a short to ground.</p> <ul style="list-style-type: none"> <li>Place one test lead on the shutdown unit Modicon™ communication bus supply wire at the SDU410 unit.</li> <li>Place the other test lead on panel ground.</li> </ul> <p>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</p> | <p>Less than 10 ohms?</p> <p><b>YES</b></p> <p><b>Repair :</b></p> <p>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15.<br/>(/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</p> | <p><b>Repair complete</b></p> |
|                                                                                                                                                                                                                                                                                                                                                                                                           | <p>Less than 10 ohms?</p> <p><b>NO</b></p>                                                                                                                                                                               | <p><b>2A</b></p>              |

## STEP 2. Check the OEM wiring harness.

### STEP 2A. Check the low speed oil pressure signal and return wires for an open.

**Conditions :**

- Disconnect the OEM wiring harness from the customer interface box at the C4 connector.
- Disconnect the OEM wiring harness at the C11 connector.

| Action | Specification/Repair | Next Step |
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| <p>Check the low speed oil pressure signal and return wires for an open.</p> <ul style="list-style-type: none"> <li>Place one test lead on the low speed oil pressure signal pin at the C4 connector.</li> <li>Place the other test lead on the low speed oil pressure signal pin at the C11 connector.</li> <li>Place one test lead on the low speed oil pressure return pin at the C4 connector.</li> <li>Place the other test lead on the low speed oil pressure return pin at the C11 connector.</li> </ul> <p>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</p> | <p>Less than 10 ohms?</p> <p><b>YES</b></p>                                                                                                                                                                             | <p><b>2A-1</b></p>            |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <p>Less than 10 ohms?</p> <p><b>NO</b></p> <p><b>Repair :</b></p> <p>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15.<br/>(/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</p> | <p><b>Repair complete</b></p> |

**STEP 2A-1. Check the low speed oil pressure signal and return wires for a wire-to-wire short.**

| <p><b>Conditions :</b></p> <ul style="list-style-type: none"> <li>Disconnect the OEM wiring harness from the customer interface box at the C4 connector.</li> <li>Disconnect the OEM wiring harness at the C11 connector.</li> </ul> |                      |           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|-----------|
| Action                                                                                                                                                                                                                               | Specification/Repair | Next Step |

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| <p>Check the low speed oil pressure signal and return wires for a wire-to-wire short.</p> <ul style="list-style-type: none"> <li>Place one test lead on the low speed oil pressure signal pin at the C4 connector.</li> <li>Place the other test lead on all other pins at the C4 connector.</li> <li>Place one test lead on the low speed oil pressure return pin at the C4 connector.</li> <li>Place the other test lead on all other pins at the C4 connector.</li> </ul> <p>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</p> | <p>Less than 10 ohms?</p> <p><b>YES</b></p> <p><b>Repair :</b></p> <p>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15.<br/>(/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</p> | <p><b>Repair complete</b></p> |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>Less than 10 ohms?</p> <p><b>NO</b></p>                                                                                                                                                                               | <p><b>2B</b></p>              |

## STEP 2B. Check the high speed oil pressure signal and return wires for an open.

### Conditions :

- Disconnect the OEM wiring harness from the customer interface box at the C4 connector.
- Disconnect the OEM wiring harness at the C11 connector.

| Action | Specification/Repair | Next Step |
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| <p>Check the high speed oil pressure signal and return wires for an open.</p> <ul style="list-style-type: none"> <li>Place one test lead on the high speed oil pressure signal pin at the C4 connector.</li> <li>Place the other test lead on the high speed oil pressure signal pin at the C11 connector.</li> <li>Place one test lead on the high speed oil pressure return pin at the C4 connector.</li> <li>Place the other test lead on the high speed oil pressure return pin at the C11 connector.</li> </ul> <p>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</p> | <p>Less than 10 ohms?</p> <p><b>YES</b></p>                                                                                                                                                                             | <p><b>2B-1</b></p>            |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <p>Less than 10 ohms?</p> <p><b>NO</b></p> <p><b>Repair :</b></p> <p>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15.<br/>(/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</p> | <p><b>Repair complete</b></p> |

**STEP 2B-1. Check the high speed oil pressure signal and return wires for a wire-to-wire short.**

|                                                                                                                                                                                                                                      |                             |                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|------------------|
| <p><b>Conditions :</b></p> <ul style="list-style-type: none"> <li>Disconnect the OEM wiring harness from the customer interface box at the C4 connector.</li> <li>Disconnect the OEM wiring harness at the C11 connector.</li> </ul> |                             |                  |
| <b>Action</b>                                                                                                                                                                                                                        | <b>Specification/Repair</b> | <b>Next Step</b> |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                          |                               |
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| <p>Check the high speed oil pressure signal and return wires for a wire-to-wire short.</p> <ul style="list-style-type: none"> <li>Place one test lead on the high speed oil pressure signal pin at the C4 connector.</li> <li>Place the other test lead on all other pins at the C4 connector.</li> <li>Place one test lead on the high speed oil pressure return pin at the C4 connector.</li> <li>Place the other test lead on all other pins at the C4 connector.</li> </ul> <p>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</p> | <p>Less than 10 ohms?</p> <p><b>YES</b></p> <p><b>Repair :</b></p> <p>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15.<br/>(/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</p> | <p><b>Repair complete</b></p> |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <p>Less than 10 ohms?</p> <p><b>NO</b></p>                                                                                                                                                                               | <p><b>2C</b></p>              |

## STEP 2C. Check the coolant pressure signal and return wires for an open.

### Conditions :

- Disconnect the OEM wiring harness from the customer interface box at the C4 connector.
- Disconnect the OEM wiring harness at the C11 connector.

| Action | Specification/Repair | Next Step |
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| <p>Check the coolant pressure signal and return wires for an open.</p> <ul style="list-style-type: none"> <li>Place one test lead on the coolant pressure signal pin at the C4 connector.</li> <li>Place the other test lead on the coolant pressure signal pin at the C11 connector.</li> <li>Place one test lead on the coolant pressure return pin at the C4 connector.</li> <li>Place the other test lead on the coolant pressure return pin at the C11 connector.</li> </ul> <p>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</p> | <p>Less than 10 ohms?</p> <p><b>YES</b></p>                                                                                                                                                                             | <p><b>2C-1</b></p>            |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <p>Less than 10 ohms?</p> <p><b>NO</b></p> <p><b>Repair :</b></p> <p>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15.<br/>(/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</p> | <p><b>Repair complete</b></p> |

**STEP 2C-1. Check the coolant pressure signal and return wires for a wire-to-wire short.**

**Conditions :**

- Disconnect the OEM wiring harness from the customer interface box at the C4 connector.
- Disconnect the OEM wiring harness at the C11 connector.

| Action | Specification/Repair | Next Step |
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| <p>Check the coolant pressure signal and return wires for a wire-to-wire short.</p> <ul style="list-style-type: none"> <li>Place one test lead on the coolant pressure signal pin at the C4 connector.</li> <li>Place the other test lead on all other pins at the C4 connector.</li> <li>Place one test lead on the coolant pressure return pin at the C4 connector.</li> <li>Place the other test lead on all other pins at the C4 connector.</li> </ul> <p>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</p> | <p>Less than 10 ohms?</p> <p><b>YES</b></p> <p><b>Repair :</b></p> <p>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15.<br/>(/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</p> | <p><b>Repair complete</b></p> |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p>Less than 10 ohms?</p> <p><b>NO</b></p>                                                                                                                                                                               | <p><b>2D</b></p>              |

## STEP 2D. Check the coolant temperature signal and return wires for an open.

### Conditions :

- Disconnect the OEM wiring harness from the customer interface box at the C4 connector.
- Disconnect the OEM wiring harness at the C11 connector.

| Action | Specification/Repair | Next Step |
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| <p>Check the coolant temperature signal and return wires for an open.</p> <ul style="list-style-type: none"> <li>Place one test lead on the coolant temperature signal pin at the C4 connector.</li> <li>Place the other test lead on the coolant temperature signal pin at the C11 connector.</li> <li>Place one test lead on the coolant temperature return pin at the C4 connector.</li> <li>Place the other test lead on the coolant temperature return pin at the C11 connector.</li> </ul> <p>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</p> | <p>Less than 10 ohms?</p> <p><b>YES</b></p>                                                                                                                                                                             | <p><b>2D-1</b></p>            |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <p>Less than 10 ohms?</p> <p><b>NO</b></p> <p><b>Repair :</b></p> <p>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15.<br/>(/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</p> | <p><b>Repair complete</b></p> |

**STEP 2D-1. Check the coolant temperature signal and return wires for a wire-to-wire short.**

**Conditions :**

- Disconnect the OEM wiring harness from the customer interface box at the C4 connector.
- Disconnect the OEM wiring harness at the C11 connector.

| Action | Specification/Repair | Next Step |
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| <p>Check the coolant temperature signal and return wires for a wire-to-wire short.</p> <ul style="list-style-type: none"> <li>• Place one test lead on the coolant temperature signal pin at the C4 connector.</li> <li>• Place the other test lead on all other pins at the C4 connector.</li> <li>• Place one test lead on the coolant temperature return pin at the C4 connector.</li> <li>• Place the other test lead on all other pins at the C4 connector.</li> </ul> <p>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</p> | <p>Less than 10 ohms?</p> <p><b>YES</b></p> <p><b>Repair :</b></p> <p>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15.<br/>(/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</p> | <p><b>Repair complete</b></p> |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <p>Less than 10 ohms?</p> <p><b>NO</b></p>                                                                                                                                                                               | <p><b>2E</b></p>              |

## STEP 2E. Check the engine speed 1 signal and return wires for an open.

### Conditions :

- Disconnect the OEM wiring harness from the customer interface box at the C4 connector.
- Disconnect the OEM wiring harness at the C11 connector.

| Action | Specification/Repair | Next Step |
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| <p>Check the engine speed 1 signal and return wires for an open.</p> <ul style="list-style-type: none"> <li>Place one test lead on the engine speed 1 signal pin at the C4 connector.</li> <li>Place the other test lead on the engine speed 1 signal pin at the C11 connector.</li> <li>Place one test lead on the engine speed 1 return pin at the C4 connector.</li> <li>Place the other test lead on the engine speed 1 return pin at the C11 connector.</li> </ul> <p>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</p> | <p>Less than 10 ohms?</p> <p><b>YES</b></p>                                                                                                                                                                             | <p><b>2E-1</b></p>            |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <p>Less than 10 ohms?</p> <p><b>NO</b></p> <p><b>Repair :</b></p> <p>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15.<br/>(/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</p> | <p><b>Repair complete</b></p> |

## STEP 2E-1. Check the engine speed 1 signal and return wires for a wire-to-wire short.

### Conditions :

- Disconnect the OEM wiring harness from the customer interface box at the C4 connector.
- Disconnect the OEM wiring harness at the C11 connector.

| Action | Specification/Repair | Next Step |
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| <p>Check the engine speed 1 signal and return wires for a wire-to-wire short.</p> <ul style="list-style-type: none"> <li>Place one test lead on the engine speed 1 signal pin at the C4 connector.</li> <li>Place the other test lead on all other pins at the C4 connector.</li> <li>Place one test lead on the engine speed 1 return pin at the C4 connector.</li> <li>Place the other test lead on all other pins at the C4 connector.</li> </ul> <p>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</p> | <p>Less than 10 ohms?</p> <p><b>YES</b></p> <p><b>Repair :</b></p> <p>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15.<br/>(/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</p> | <p><b>Repair complete</b></p> |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <p>Less than 10 ohms?</p> <p><b>NO</b></p>                                                                                                                                                                               | <p><b>2F</b></p>              |

## STEP 2F. Check the engine speed 2 signal and return wires for an open.

### Conditions :

- Disconnect the OEM wiring harness from the customer interface box at the C4 connector.
- Disconnect the OEM wiring harness at the C11 connector.

| Action | Specification/Repair | Next Step |
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| <p>Check the engine speed 2 signal and return wires for an open.</p> <ul style="list-style-type: none"> <li>Place one test lead on the engine speed 2 signal pin at the C4 connector.</li> <li>Place the other test lead on the engine speed 2 signal pin at the C11 connector.</li> <li>Place one test lead on the engine speed 2 return pin at the C4 connector.</li> <li>Place the other test lead on the engine speed 2 return pin at the C11 connector.</li> </ul> <p>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</p> | <p>Less than 10 ohms?</p> <p><b>YES</b></p>                                                                                                                                                                             | <p><b>2F-1</b></p>            |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <p>Less than 10 ohms?</p> <p><b>NO</b></p> <p><b>Repair :</b></p> <p>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15.<br/>(/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</p> | <p><b>Repair complete</b></p> |

## STEP 2F-1. Check the engine speed 2 signal and return wires for a wire-to-wire short.

### Conditions :

- Disconnect the OEM wiring harness from the customer interface box at the C4 connector.
- Disconnect the OEM wiring harness at the C11 connector.

| Action | Specification/Repair | Next Step |
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| <p>Check the engine speed 2 signal and return wires for a wire-to-wire short.</p> <ul style="list-style-type: none"> <li>• Place one test lead on the engine speed 2 signal pin at the C4 connector.</li> <li>• Place the other test lead on all other pins at the C4 connector.</li> <li>• Place one test lead on the engine speed 2 return pin at the C4 connector.</li> <li>• Place the other test lead on all other pins at the C4 connector.</li> </ul> <p>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</p> | <p>Less than 10 ohms?</p> <p><b>YES</b></p> <p><b>Repair :</b></p> <p>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15.<br/>(/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</p> | <p><b>Repair complete</b></p>                               |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>Less than 10 ohms?</p> <p><b>NO</b></p>                                                                                                                                                                               | <p><b>Contact a Cummins® Authorized Repair Location</b></p> |